Cloud Computing – III UNIT

2 marks:

1. What is Aneka?

Aneka is Manjrasoft's solution for developing, deploying, and managing Cloud applications. It consists of a scalable Cloud middleware that can be deployed on top of heterogeneous computing resources. It offers an extensible collection of services coordinating the execution of applications, helping administrators to monitor the status of the Cloud, and providing integration with existing Cloud technologies.

2. What is Aneka container?

The Aneka container constitutes the building block of Aneka Clouds and represents the runtime machinery available to services and applications. The container is the unit of deployment in Aneka Clouds, and it is a lightweight software layer designed to host services and interact with the underlying operating system and hardware. The main role of the container is to provide a lightweight environment where to deploy services.

3. List the services installed in the Aneka container.

• Fabric services • Foundation services • Application services

4. Expand PAL & CLI.

PAL - Platform Abstraction Layer
CLI - Common Language Infrastructure

5. What is Platform Abstraction Layer (PAL)?

The Platform Abstraction Layer (PAL) provides the container with a uniform interface for accessing the relevant hardware and operating system information, thus allowing the rest of the container to run unmodified on any platform supported. PAL gives uniform access to the hosting platform's extended and additional properties.

6. What is Common Language Infrastructure (CLI)?

The Common Language Infrastructure (CLI), which is the specification introduced in the ECMA-334 of standard defines a common runtime environment and application model for executing programs, but does not provide any interface to access the hardware or to collect performance data from the hosting operating system.

7. List the services that are in charge of managing resources.

Index Service (or Membership Catalogue), Reservation Service, and Resource Provisioning Service

8. What is reporting service?

The Reporting Service manages the store for monitored data and makes them accessible to other services or external applications for analysis purposes. On each node, an instance of the Monitoring Service acts as a gateway to the Reporting Service and forwards all the monitored data that has been collected on the node to it. Any service wanting to publish monitoring data can leverage the local monitoring service without knowing the details of the entire infrastructure.

9. What are the 3 nodes used by the logical organization?

Master node, Worker node and Storage node

10. Which services are hosted in Master node.

- Index Service (master copy)
- Heartbeat Service
- Logging Service
- Reservation Service
- Resource Provisioning Service
- Accounting Service

11. What are the different models supports by the Aneka clouds?

- Task Model
- Thread Model
- MapReduce Model
- Parameter Sweep Model

12. What is Aneka daemon?

The infrastructure is deployed by harnessing a collection of nodes and installing on them the Aneka node manager, also called the Aneka daemon. The daemon constitutes the remote management service used to deploy and control container instances. The collection of resulting containers identifies the Aneka Cloud.

13. What is Aneka repository?

Aneka repository provides storage for all the libraries required to lay out and install the basic Aneka platform. These libraries constitute the software image for the node manager and the container programs. Repositories can make libraries available through a variety of communication channels, **such as HTTP, FTP, common file sharing, and so on.**

14. What is private cloud deployment mode?

A private deployment mode is mostly constituted by local physical resources and infrastructure management software providing access to a local pool of nodes, which might be virtualized. In this scenario Aneka Clouds are created by harnessing a heterogeneous pool of resources such has desktop machines, clusters, or workstations.

15. What is public cloud deployment mode?

Public Cloud deployment mode features the installation of Aneka master and worker nodes over a completely virtualized infrastructure that is hosted on the infrastructure of one or more resource providers. In this case it is possible to have a static deployment where the nodes are provisioned beforehand and used as though they were real machines.

16. What is hybrid cloud deployment mode?

The hybrid deployment model constitutes the most common deployment of Aneka. In many cases, there is an existing computing infrastructure that can be leveraged to address the computing needs of applications. This scenario constitutes the most complete deployment for Aneka that is able to leverage all the capabilities of the framework.

17. What are the categories of application models?

(i) Applications whose tasks are generated by the user
(ii) Applications whose tasks are generated by the runtime infrastructure

18. What is service model?

The Aneka Service Model defines the basic requirements to implement a service that can be hosted in an Aneka Cloud. The container defines the runtime environment in which services are hosted. Each service that is hosted in the container must be compliant with the IService interface which exposes Name and status, Control operations, Message handling.

19. Mention different features of Aneka management tools.

- Infrastructure management
- Platform management
- Application management

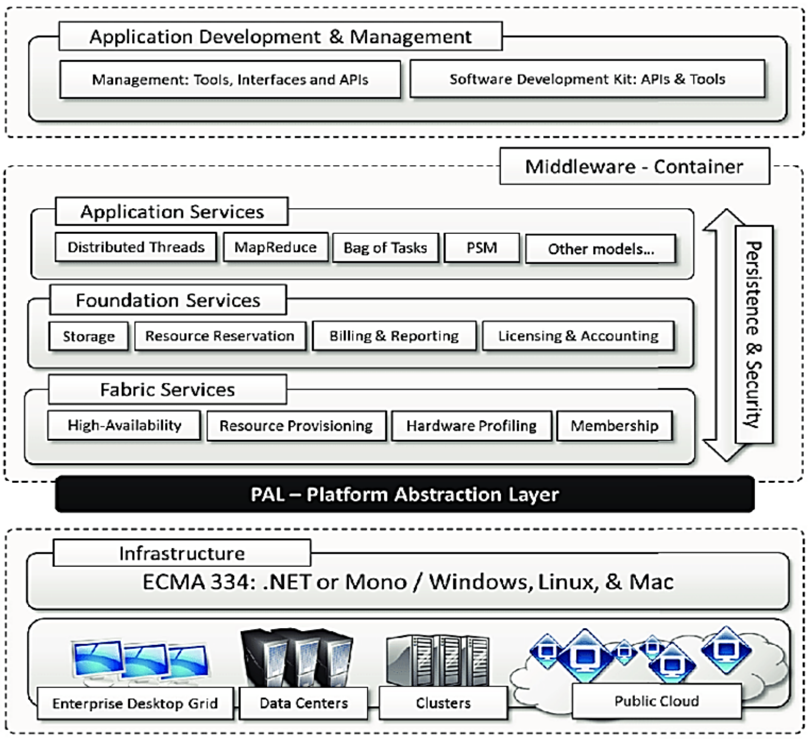
20. What is infrastructure management?

Aneka leverages virtual and physical hardware in order to deploy Aneka Clouds. Virtual hardware is generally managed by means of the Resource Provisioning Service, which acquires resources on demand according to the need of applications, while physical hardware is directly managed by the Administrative Console by leveraging the Aneka management API of the PAL. The management features are mostly concerned with the provisioning of physical hardware and the remote installation of Aneka on the hardware.

Long answer questions

1. Explain different components of Aneka framework.

Aneka is a software platform for developing Cloud computing applications. According to the Cloud Reference Model , Aneka is a Pure PaaS solution for Cloud computing. The framework provides both a middleware for managing and scaling distributing applications and an extensible set of APIs for developing them.



The core infrastructure of the system provides a uniform layer allowing the framework to be deployed over different platform and operating systems.

The physical and virtual resources representing the bare metal of the Cloud are managed by the Aneka container, which is installed on each node and constitutes the basic building block of the middleware.

A collection of interconnected containers constitute the Aneka Cloud: a single domain in which services are made available to users and developers.

The container features three different classes of services: Fabric Services, Foundation Services, and Execution Services. These respectively take care of infrastructure management, supporting services for the Cloud, and application management and execution. These services are made available to developers and administrators by the means of the application management and development layer, which includes interfaces and APIs for developing Cloud applications, and the management tools and interfaces for controlling Aneka Clouds.

2. Explain Fabric Services offered by Aneka Cloud Platform.

Fabric services define the lowest level of the software stack representing the Aneka Container.

(i) Profiling and Monitoring: Profiling and monitoring services are mostly exposed through the Heartbeat, Monitoring, and Reporting services.

The ***Heartbeat service*** periodically collects the dynamic performance information about the node, and publishes this information to membership service in the Aneka Cloud. These data are collected by the index node of the Cloud. The information published by the Heartbeat service is mostly concerned with the properties of the node.

The ***Reporting Service*** manages the store for monitored data and makes them accessible to other services or external applications for analysis purposes.

On each node, an instance of the ***Monitoring Service*** acts as a gateway to the Reporting Service and forwards all the monitored data that has been collected on the node to it.

(ii) Resource Management: Aneka provides a collection of services that are in charge of managing resources. These are: Index Service (or Membership Cata- logue), Reservation Service, and Resource Provisioning Service.

The ***Membership catalogue*** is the fundamental component for resource management since it keeps track of the basic node information for all the nodes that are connected or disconnected.

Dynamic resource provisioning allows the integration and the management, of virtual resources leased from IaaS providers into the Aneka Cloud. Aneka defines a very flexible infrastructure for resource provisioning where it is possible to change the logic that triggers provisioning, support several back-ends, and change the runtime strategy.

Resource provisioning is a feature designed to support Quality of Service (QoS) requirements driven execution of applications. Therefore, it mostly serves requests coming from the ***Reservation Service*** or the Scheduling services.

3. Write a note of Platform abstraction layer.

The Platform Abstraction Layer (PAL) provides the container with a uniform interface for accessing the relevant hardware and operating system information, thus allowing the rest of the container to run unmodified on any platform supported. The PAL is responsible for detecting the supported hosting environment, and providing the corresponding to interact with it for supporting the activity of the container.

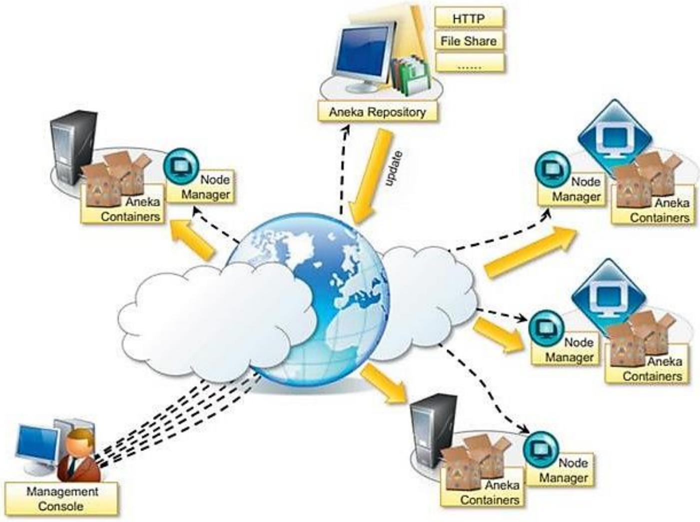
It provides the following features:

• Uniform and platform-independent implementation interface for accessing the hosting platform.
• Uniform access to extended and additional properties of the hosting platform
• Uniform and platform independent access to remote nodes .
• Uniform and platform independent management interfaces.

The PAL is a small layer of software comprising a detection engine which automatically configures the container at boot time with the platform-specific component to access the above information, and an implementation of the abstraction layer for the Windows, Linux, and Mac OS X operating system. The collectible data that are exposed by the PAL are the following:

• Number of cores, frequency, and CPU usage
• Memory size and usage
• Aggregate available disk space.
• Network addresses and devices attached to the node

4. Explain the infrastructure organization of Aneka clouds.



The scenario is a reference model for all the different deployments Aneka supports.

A central role is played by the ***Administrative Console***, which performs all the required management operations.

A fundamental element for Aneka Cloud deployment is constituted by ***repositories***. A repository provides storage for all the libraries required to lay out and install the basic Aneka platform. These libraries constitute the software image for the node manager and the container programs. Repositories can make libraries available through a variety of communication channels, such as HTTP, FTP, common file sharing, and so on.

The ***Management Console*** can manage multiple repositories and select the one that best suits the specific deployment.

The infrastructure is deployed by harnessing a collection of nodes and installing on them the Aneka node manager, also called the ***Aneka daemon***. The daemon constitutes the remote management service used to deploy and control container instances. The collection of resulting containers identifies the Aneka Cloud

From an infrastructure point of view, the management of physical or virtual nodes is performed uniformly as long as it is possible to have an Internet connection and remote administrative access to the node.

5. Explain briefly about logical organization of Aneka clouds.

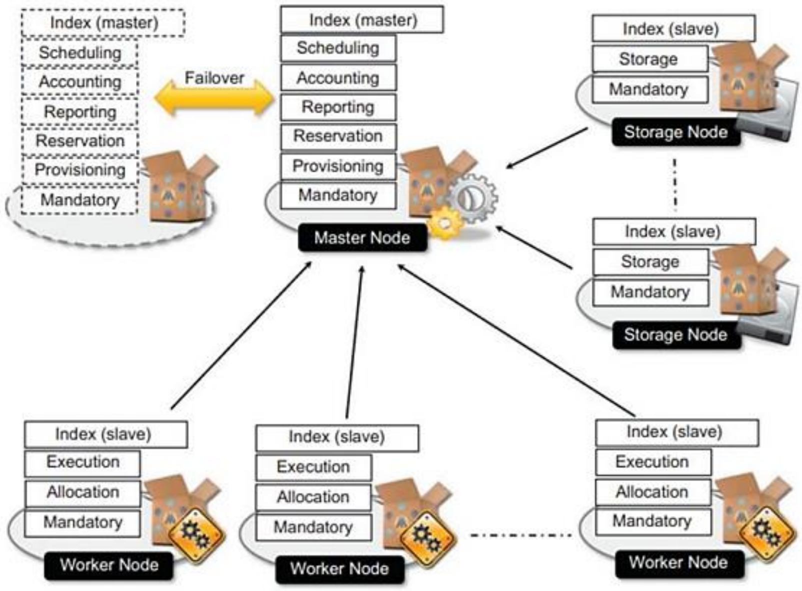
The logical organization of Aneka Clouds can be very diverse, since it strongly depends on the configuration selected for each of the container instances belonging to the Cloud. The most common scenario is to use a master-worker configuration with separate nodes for storage.

The ***master node*** features all the services that are most likely to be present in one single copy and that provide the intelligence of the Aneka Cloud.

A common configuration of the master node is as follows:

• Index Service (master copy) • Heartbeat Service • Logging Service
• Reservation Service • Resource Provisioning Service • Accounting Service

The master node also provides connection to an RDBMS facility where the state of several services is maintained. For the same reason, all the scheduling services are maintained in the master node.



The ***worker nodes*** constitute the workforce of the Aneka Cloud and are generally configured for the execution of applications. They feature the mandatory services and the specific execution services of each of the supported programming models in the Cloud.

A very common configuration is the following:

• Index Service • Heartbeat Service • Logging Service • Allocation Service
• Monitoring Service • Execution Services for the supported programming models

Storage nodes are optimized to provide storage support to applications. They feature, among the mandatory and usual services, the presence of the Storage Service. The number of storage nodes strictly depends on the predicted workload and storage consumption of applications.

The common configuration of a storage node is the following:
• Index Service • Heartbeat Service • Logging Service • Monitoring Service • Storage Service

6. Differentiate private, public and hybrid cloud deployment mode.

Difference	Public Cloud	Private Cloud	Hybrid Cloud
Data Tenancy	**Multi-tenancy:** The data of numerous companies is stored in a shared environment	**Single Tenancy:** The data of only a single organization is stored in the cloud	The data stored in the public cloud is shared, and the data stored in the private cloud is not shared

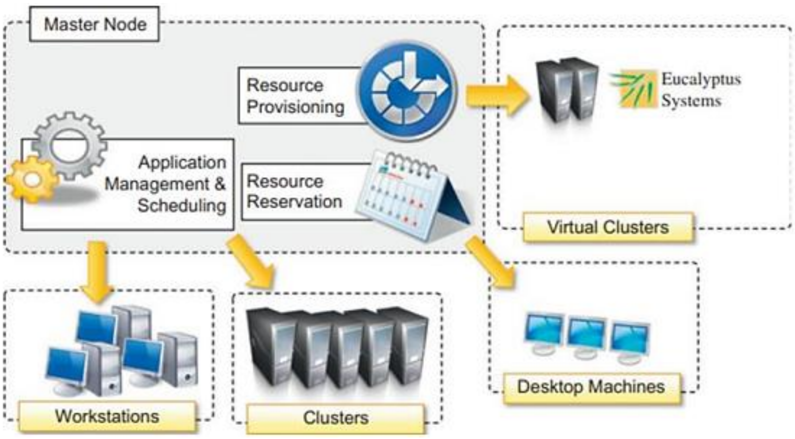
Cloud Services	Open to public	Only that specific organization can use the cloud services	Services on the public cloud can be accessed by everyone, whereas services in the private cloud can be accessed only by that organization
Connectivity	Over the internet	Over the organization's private network	Over the internet for public cloud services and organization's private network for private cloud services
Costs	Less expensive as the cloud service provider offers all the resources	Very expensive as the organization has to purchase all the resources	Less costly for public cloud and more expensive for private cloud resources
Scalability and Flexibility	High	High	High
Security	Low	High	Public Cloud – Low Private Cloud – High

7. Explain private cloud deployment mode.

A private deployment mode is mostly constituted by local physical resources and infrastructure management software providing access to a local pool of nodes, which might be virtualized. In this scenario Aneka Clouds are created by harnessing a heterogeneous pool of resources such has desktop machines, clusters, or workstations. These resources can be partitioned into different groups, and Aneka can be configured to leverage these resources according to application needs.

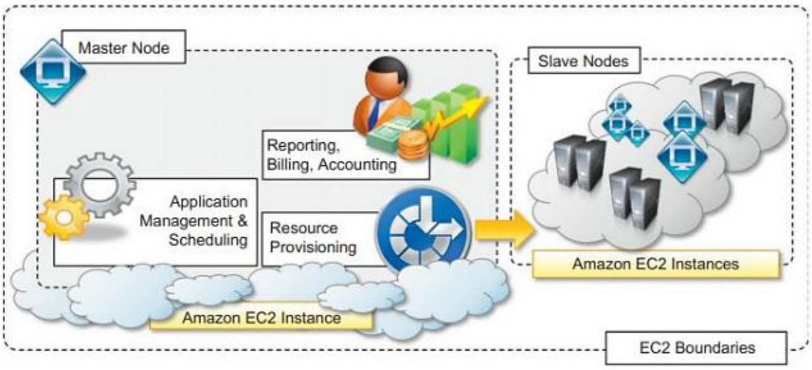
This deployment is acceptable for a scenario in which the workload of the system is predictable and a local virtual machine manager can easily address excess capacity demand. Most of the Aneka nodes are constituted of physical nodes with a long lifetime and a static configuration and generally do not need to be reconfigured often.

The different nature of the machines harnessed in a private environment allows for specific policies on resource management and usage that can be accomplished by means of the Reservation Service. For example, desktop machines that are used during the day for office automation can be exploited outside the standard working hours to execute distributed applications.



8. Explain public cloud deployment mode.

Public Cloud deployment mode features the installation of Aneka master and worker nodes over a completely virtualized infrastructure that is hosted on the infrastructure of one or more resource providers. In this case it is possible to have a static deployment where the nodes are provisioned beforehand and used as though they were real machines.



The deployment is generally contained within the infrastructure boundaries of a single IaaS provider. The reasons for this are to minimize the data transfer between different providers, which is generally priced at a higher cost, and to have better network performance.

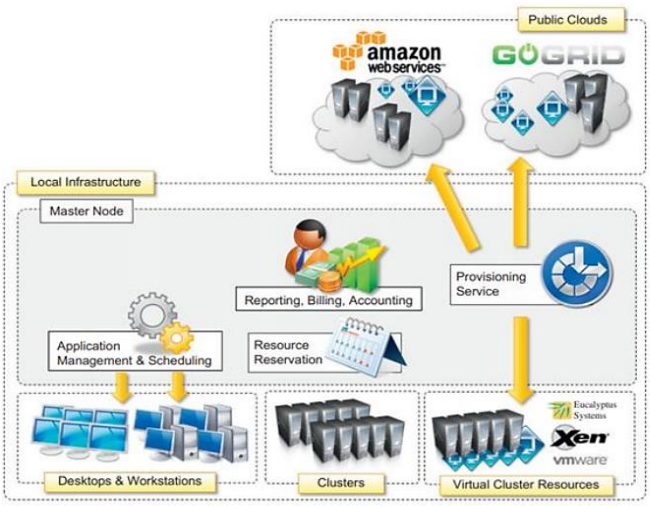
In this scenario it is possible to deploy an Aneka Cloud composed of only one node and to completely leverage dynamic provisioning to elastically scale the infrastructure on demand. A fundamental role is played by the Resource Provisioning Service, which can be configured with different images and templates to instantiate. Other important services that have to be included in the master node are the Accounting and Reporting Services.

9. Explain hybrid cloud deployment mode.

The hybrid deployment model constitutes the most common deployment of Aneka. In many cases, there is an existing computing infrastructure that can be leveraged to address the computing needs of applications. This scenario constitutes the most complete deployment for Aneka that is able to leverage all the capabilities of the framework:

• Dynamic Resource Provisioning
• Resource Reservation
• Workload Partitioning
• Accounting, Monitoring, and Reporting

In a hybrid scenario, heterogeneous resources can be used for different purposes. As we discussed in the case of a private cloud deployment, desktop machines can be reserved for low priority workload outside the common working hours. The majority of the applications will be executed on workstations and clusters, which are the nodes that are constantly connected to the Aneka Cloud. Any additional computing capability demand can be primarily addressed by the local virtualization facilities, and if more computing power is required, it is possible to leverage external IaaS providers.



10.Explain Aneka SDK.

Aneka provides APIs for developing applications on top of existing programming models, implementing new programming models, and developing new services to integrate into the Aneka Cloud. The SDK provides support for both programming models and services by means of the Application Model and the Service Model.

(i) Application Model: The Application Model represents the minimum set of APIs that is common to all the programming models for representing and programming distributed applications on top of Aneka. Each distributed application running on top of Aneka is an instance of the ApplicationBase <M> class, where M identifies the specific type of application manager used to control the application.
Aneka further specializes applications into two main categories:

(1) Applications whose tasks are generated by the user
(2) Applications whose tasks are generated by the runtime infrastructure

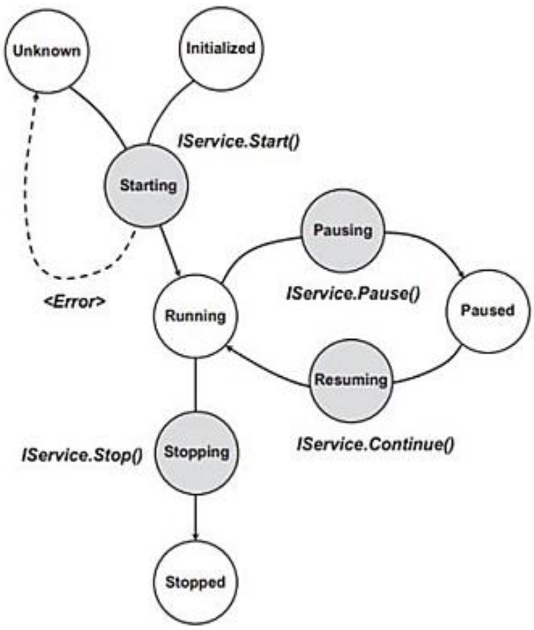
The first category is the most common and it is used as a reference for several programming models supported by Aneka: the Task Model, the Thread Model, and the Parameter Sweep Model. Applications that fall into this category are composed of a collection of units of work submitted by the user and represented by the WorkUnit class.

The second category covers the case of MapReduce and all those other scenarios in which the units of work are generated by the runtime infrastructure rather than the user.

Table 5.1 Aneka's Application Model Features

Category	Description	Base Application Type	Work Units?	Programming Models
Manual	Units of work are generated by the user and submitted through the application.	*AnekaApplication <W,M>* *IManualApplicationManager <W>* *ManualApplicationManager <W>*	Yes	Task Model Thread Model Parameter Sweep Model
Auto	Units of work are generated by the runtime infrastructure and managed internally.	*ApplicationBase <M>* *IAutoApplicationManager*	No	*MapReduce* Model

(ii) Service Model: The Aneka Service Model defines the basic requirements to implement a service that can be hosted in an Aneka Cloud. The container defines the runtime environment in which services are hosted. Each service that is hosted in the container must be compliant with the IService interface which exposes Name and status, Control operations, Message handling.



This figure describes the reference life cycle of each service instance in the Aneka container. The shaded balloons indicate transient states; the white balloons indicate steady states.

A service instance can initially be in the ***Unknown*** or ***Initialized*** state, a condition that refers to the creation of the service instance. Once the container is started, it will iteratively call the ***Start*** method on each service method. As a result, the service instance is expected to be in a ***Starting*** state until the startup process is completed, after which it will exhibit the ***Running*** state. This is the only state in which the service is able to process messages. If an exception occurs, the service will fall back to the Unknown state, thus signalling an error.

When a service is running it is possible to pause its activity by calling the ***Pause*** method and resume it by calling ***Continue***. As described in the figure, the service moves first into the ***Pausing*** state, thus reaching the ***Paused*** state. From this state, it moves into the ***Resuming*** state while restoring its activity to return to the Running state.

When the container shuts down, the ***Stop*** method is iteratively called on each service running, and services move first into the transient ***Stopping*** state to reach the final ***Stopped*** state, where all resources that were initially allocated have been released.

11. Write a node on Application model of Aneka SDK.

Refer Question 10

12. Explain service life cycle of Aneka service model.

Refer Question 10

13. Explain the different categories of management tools.

Aneka is a pure PaaS implementation and requires virtual or physical hardware to be deployed. Hence, infrastructure management, together with facilities for installing logical clouds on such infrastructure, is a fundamental feature of Aneka's management layer. This layer also includes capabilities for managing services and applications running in the Aneka Cloud

(i) ***Infrastructure management:*** Aneka leverages virtual and physical hardware in order to deploy Aneka Clouds. Virtual hardware is generally managed by means of the Resource Provisioning Service, which acquires resources on demand according to the need of applications, while physical hardware is directly managed by the Administrative Console by leveraging the Aneka management API of the PAL. The management features are mostly concerned with the provisioning of physical hardware and the remote installation of Aneka on the hardware.

(ii) ***Platform management***: A collection of connected containers defines the platform on top of which applications are executed. The features available for platform management are mostly concerned with the logical organization and structure of Aneka Clouds. It is possible to partition the available hardware into several Clouds variably configured for different purposes. Services implement the core features of Aneka Clouds and the management layer exposes operations for some of them, such as Cloud monitoring, resource provisioning and reservation, user management, and application profiling.

(iii) ***Application management***: Applications identify the user contribution to the Cloud. The management APIs provide administrators with monitoring and profiling features that help them track the usage of resources and relate them to users and applications. This is an important feature in a cloud computing scenario in which users are billed for their resource usage. Aneka exposes capabilities for giving summary and detailed information about application execution and resource utilization